

Developmental and Epileptic Encephalopathy 55 (DEE55 / PIGP Deficiency): A Comprehensive Disease Characteristics Report

Summary

Developmental and Epileptic Encephalopathy 55 (DEE55; OMIM #617599) is an ultra-rare, autosomal-recessive **inherited glycosylphosphatidylinositol (GPI) deficiency (IGD)** caused by biallelic loss-of-function variants in **PIGP** (Phosphatidylinositol Glycan Anchor Biosynthesis, Class P; OMIM *605938; HGNC:3046; formerly *DSCR5*; located at 21q22.13). PIGP encodes an accessory subunit of the endoplasmic-reticulum **GPI-GlcNAc transferase** complex, the enzyme that catalyzes the committed first step of GPI anchor biosynthesis. When PIGP function is reduced, the cell produces fewer GPI anchors, which in turn lowers the cell-surface expression of the ≥ 150 human GPI-anchored proteins (GPI-APs) that are essential for normal neuronal development and function. The downstream deficit of these surface proteins produces the disease's cardinal triad: **early-onset refractory seizures, muscular hypotonia, and profound global developmental delay / intellectual disability**, typically accompanied by cerebral and cerebellar atrophy on neuroimaging.

The disease was first defined molecularly in 2017 by Johnstone and colleagues in two compound-heterozygous siblings, and remained described in only ~7 patients as of 2023. Since then the recognized phenotypic spectrum has broadened substantially — from the classic DEE presentation, to **Multiple Congenital Anomalies–Hypotonia–Seizures Syndrome (MCAHS)**, and at the most severe end to **prenatal lethality** (a fetus terminated at 17 weeks gestation). Because complete GPI deficiency is embryonic-lethal, all viable patients carry **hypomorphic (partial-loss-of-function) biallelic variants**. A recurrent frameshift allele, **c.456delA (p.Glu153Asnfs*34)**, has been observed both in compound-heterozygous and homozygous states across unrelated families and is ultra-rare in population databases (≈ 9 heterozygous carriers, 0 homozygotes in gnomAD).

There is **no curative or disease-modifying therapy**. Management is symptomatic and multidisciplinary — antiseizure medications for the refractory epilepsy, plus supportive care (feeding-tube nutrition, physical/occupational/speech therapy). Empiric nutritional interventions such as **pyridoxine (vitamin B6)** and the **ketogenic diet** are used in the broader GPI-anchor-disorder group. Prognosis is severe, with profound lifelong disability and substantial mortality ($\approx 18\%$ deceased across the largest IGD cohort). Diagnosis is achieved by whole-exome or whole-genome sequencing — notably, targeted epilepsy gene panels can miss PIGP — supported functionally by flow-cytometric detection of reduced GPI-anchored proteins on blood cells.

Key Findings

Finding 1 — DEE55 is caused by biallelic PIGP variants and is an autosomal-recessive inherited GPI deficiency

The molecular basis of DEE55 was established by **Johnstone et al. 2017** [PMID: 28334793](#), who reported two siblings with compound-heterozygous variants in PIGP (NM_153681.2: **c.74T>C**, **p.Met25Thr** and **c.456delA**, **p.Glu153AsnFs*34**). As the authors state: *"Here, we report two siblings with compound heterozygous variants in the gene phosphatidylinositol glycan anchor biosynthesis, class P (PIGP) (NM_153681.2: c.74T > C;p.Met25Thr and c.456delA;p.Glu153AsnFs*34). PIGP encodes a subunit of the enzyme that catalyzes the first step of GPI anchor biosynthesis."*

Functional work confirmed the loss-of-function mechanism: *"Functional studies with patient cells showed reduced PIGP mRNA levels, and an associated reduction of GPI-anchored cell surface proteins, which was rescued by exogenous expression of wild-type PIGP. This work associates mutations in the PIGP gene with a novel autosomal recessive IGD."* The rescue by wild-type PIGP demonstrates that the surface-protein deficit is a direct consequence of PIGP dysfunction, satisfying a key criterion for causality. The disease is thus an **autosomal-recessive Mendelian disorder** within the family of inherited GPI deficiencies (a subgroup of the congenital disorders of glycosylation).

Finding 2 — Core clinical phenotype: early-onset refractory seizures, hypotonia, and profound developmental delay

Both index PIGP patients *"presented with early-onset refractory seizures, hypotonia, and profound global developmental delay, reminiscent of other IGD phenotypes"* (Johnstone 2017, PMID: 28334793). The frequency and character of these features are well quantified in the largest IGD cohort to date, Sidpra et al. 2024 (PMID: 38456468, n = 83), which reported: *"Core clinical features were developmental delay or intellectual disability (DD/ID, 90%), seizures (83%), hypotonia (72%) and motor symptoms (64%)."* Seizure onset is typically in infancy: *"Median age at seizure onset was 6 months."*

Neuroimaging in the IGD group commonly shows cerebral atrophy (75%), cerebellar atrophy (60%), corpus-callosum anomalies (57%), and a distinctive symmetric restricted diffusion of the central tegmental tracts (60%). The OMIM clinical synopsis for DEE55 specifically (per Vetro et al. 2020, curated in OMIM #617599) describes refractory seizures beginning in the first weeks-to-months of life, profound intellectual disability, absent speech, spastic quadriplegia, dyskinetic movements, cortical visual impairment, and feeding-tube dependence, with MRI showing cerebral atrophy, a thin corpus callosum, and abnormal white-matter signal.

Suggested HPO terms: Seizure (HP:0001250); Infantile spasms / early-onset epilepsy (HP:0012469 / HP:0011097); Muscular hypotonia (HP:0001252); Global developmental delay (HP:0001263); Profound intellectual disability (HP:0002187); Absent speech (HP:0001344); Spastic tetraplegia (HP:0002510); Cortical visual impairment (HP:0100704); Cerebral atrophy (HP:0002059); Cerebellar atrophy (HP:0001272); Thin corpus callosum (HP:0033725).

Finding 3 — Phenotypic spectrum extends to MCAHS and prenatal lethality

Martín-Grau et al. 2023 (PMID: 37125481) expanded the PIGP phenotype beyond classic DEE. They described two compound-heterozygous siblings carrying PIGP NM_153682.3:c.2T>C (p.?) and a 136-kb deletion at 21q22.13 (GRCh37 chr21:38,329,939–38,466,066) removing the entire PIGP gene: *"Both were compound heterozygous of pathogenic variants in PIGP gene: NM_153682.3:c.2 T > C(p.?) and a 136 Kb deletion ... affecting the entire PIGP gene."* One child had dysmorphic features, congenital anomalies, hypotonia, and epileptic encephalopathy; the other was *"a fetus with a severe malformation disorder at 17 weeks of gestation whose pregnancy was interrupted."*

The authors emphasized the rarity of the condition — *"To date, the PIGP gene has only been related to Developmental and Epileptic Encephalopathy 55 (MIM#617599) in just seven patients"* — and proposed a nosological expansion: *"Our results extend the clinical phenotype associated*

to *PIGP* gene and propose to include it as a novel cause of Multiple Congenital Anomalies-Hypotonia-Seizures syndrome." This establishes a **severity continuum** from prenatal-lethal malformation through MCAHS to the classic postnatal DEE presentation, consistent with variable expressivity driven by the residual GPI-biosynthetic activity of each allele combination.

Finding 4 — Mechanism: PIGP is a subunit of the ER GPI-GlcNAc transferase catalyzing the committed first step of GPI biosynthesis

"PIGP encodes a subunit of the enzyme that catalyzes the first step of GPI anchor biosynthesis" (Johnstone 2017, [PMID: 28334793](#)). This first step — transfer of N-acetylglucosamine (GlcNAc) from UDP-GlcNAc to phosphatidylinositol to form GlcNAc-PI on the cytoplasmic face of the ER — is the committed, rate-defining reaction of the entire pathway. Downstream, the GPI anchor is assembled and transferred to hundreds of proteins: *"At least 150 human proteins are glycosylphosphatidylinositol-anchored proteins (GPI-APs)"* (Kinoshita review, [PMID: 32156170](#); an updated review, [PMID: 39129667](#), cites ≥ 160). GPI attachment is required for the cell-surface expression of these proteins.

A critical mechanistic constraint explains why patients survive at all: *"Patients with IGD have only a partial deficiency because complete GPI deficiency causes embryonic death"* (Murakami & Kinoshita 2015, [PMID: 25803904](#)). Thus all DEE55 alleles must be **hypomorphic** rather than null on both copies; the degree of residual activity, together with which step of the pathway is impaired, tunes disease severity.

Suggested GO terms: GPI anchor biosynthetic process (GO:0006506); N-acetylglucosaminyltransferase activity (GO:0016262/related); attachment of GPI anchor to protein (GO:0016255); endoplasmic reticulum membrane (GO:0005789).

Finding 5 — PIGP is one of seven components of the GPI-GlcNAc transferase complex

Martín-Grau 2023 ([PMID: 37125481](#)) situates PIGP within its multi-subunit enzyme: *"the initial stage of their biosynthesis is mediated by PIGA, PIGC, PIGH, PIGP, PIGQ, PIGY, and DMP2 genes, which have been linked to a wide spectrum of phenotypes depending on the gene damaged."* PIGA is the catalytic subunit, while PIGP is an accessory/regulatory subunit (DPM2/DMP2 is shared with the dolichol-phosphate-mannose synthase machinery). Defects in the different subunits produce overlapping yet gene-specific IGD phenotypes; PIGA-, PIGQ-, and PIGY-related

disorders overlap clinically with PIGP-related MCAHS. This complex-level view explains the phenotypic convergence of the whole GPI-GnT subgroup on neurodevelopmental epileptic encephalopathy.

Finding 6 — Recurrent c.456delA allele; ultra-rare in population databases; standard epilepsy panels can miss PIGP

Krenn et al. 2019 (PMID: 31139695; *Ann Clin Transl Neurol* 6(5):968–973) described a second, independent family: a 2-year-old girl of unrelated Polish parents who was **homozygous for PIGP c.456delA (p.Glu153Asnfs*34)** — the very same frameshift allele carried in compound-heterozygous state by the original Johnstone siblings. This makes c.456delA a **recurrent loss-of-function allele** across the small known patient population. The variant is ultra-rare: 6 heterozygotes and no homozygotes in an in-house 16,000-exome database, and 9 heterozygous / 0 homozygous carriers in gnomAD.

Two diagnostic lessons emerge. First, **whole-exome sequencing was required**: a 100-gene epileptic-encephalopathy panel (which included other PIG genes — PIGA, PIGG, PIGN, PIGT — but not PIGP) failed to detect the diagnosis. Second, **flow cytometry of patient lymphocytes showed decreased surface GPI-anchored proteins**, providing functional confirmation. GeneMatcher yielded no additional families at the time, underscoring the disease's extreme rarity.

Finding 7 — Diagnostic approach: WES/WGS + flow cytometry of GPI-anchored proteins; variable hyperphosphatasia

Diagnosis combines genomic sequencing with a functional cell-surface assay. Murakami & Kinoshita 2015 (PMID: 25803904) note that "*Flow cytometric analysis of GPI-APs on granulocytes is also useful for the detection of IGD*" and that "*The presence of hyperphosphatasia is strong evidence of IGD.*" Importantly, hyperphosphatasia (elevated tissue-nonspecific alkaline phosphatase, a GPI-anchored protein) is chiefly associated with **late-pathway** defects (e.g., Mabry syndrome / hyperphosphatasia-with-mental-retardation from PIGV/PIGO/PGAP defects) and is **not a consistent feature of PIGP**, an early-pathway defect. Multicolor flow cytometry using markers such as FLAER, CD16, CD24, CD55, CD59 (and the T5 antibody for free GPI) yields gene/complex-specific profiles (Knaus 2019, PMID: 31353022): "*Using multicolor flow cytometry, we determined a characteristic profile for GPI transamidase deficiency.*" In PIGP cases specifically, diagnosis was made by WES

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and a whole-gene 136-kb deletion was detectable as a copy-number variant on genomic analysis

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Finding 8 — Treatment is symptomatic; pyridoxine and ketogenic diet are reported supportive options

There is no curative therapy. Management is anticonvulsant-based seizure control plus multidisciplinary supportive care. **Boyer, Johnsen & Morava 2022** (PMID: 35562242) reviewed nutritional interventions across the >160 congenital disorders of glycosylation, noting that specific therapies exist for very few subtypes: *"Specific nutritional treatment options for certain CDG types include oral supplementation of monosaccharide sugars, manganese, uridine, or pyridoxine."* They explicitly address the GPI-anchor subgroup: *"We review the dietary management in CDG with a focus on two subgroups: N-linked glycosylation defects and GPI-anchor disorders."* Pyridoxine (vitamin B6) and the ketogenic diet are the empiric interventions most often cited for GPI-anchor disorders, though robust efficacy data specific to PIGP are lacking.

Finding 9 — Severe prognosis, including prenatal lethality and ≈18% mortality across the IGD group

DEE55 carries a severe prognosis. Only ~7 PIGP patients were reported by 2023 (PMID: 37125481), and one allele combination produced fetal termination at 17 weeks. Across the largest IGD cohort (Sidpra 2024, PMID: 38456468, n = 83), *"Follow-up data were available for all individuals, 15 of whom were deceased at the time of writing"* (≈18% mortality). Morbidity is multisystemic: *"Sixty-one individuals had multisystem involvement including gastrointestinal (66%), cardiac (19%) and renal (14%) anomalies."* The OMIM synopsis notes death in childhood may occur in DEE55 specifically.

Finding 10 — Authoritative identifiers

Resource	Identifier
Disease (phenotype)	OMIM #617599 — Developmental and Epileptic Encephalopathy 55
Orphanet	PIGP-related early infantile epileptic encephalopathy (within Orphanet IGD classifications; ORPHA:1934 grouping)
Gene	PIGP — OMIM *605938; HGNC:3046 ; NCBI Gene 51227 ; Ensembl ENSG00000185808 ; UniProt P57054
Locus	21q22.13 (GRCh38 chr21:37,065,364–37,073,071) — within the Down syndrome critical region
Former/alias symbols	DSCR5 (Down syndrome critical region gene 5); DCRC; DSRC
MONDO	Align to the MONDO term for developmental and epileptic encephalopathy 55 (map to OMIM:617599)

Detailed Section-by-Section Report

1. Disease Information

DEE55 is a Mendelian, autosomal-recessive developmental and epileptic encephalopathy — a severe neurodevelopmental disorder in which epileptic activity itself is thought to contribute to developmental impairment, superimposed on the direct developmental effect of the underlying metabolic lesion. It belongs to the **inherited GPI deficiencies (IGDs)**, a subclass of the **congenital disorders of glycosylation (CDG)**. Key identifiers are listed in Finding 10. Synonyms include *PIGP-related developmental and epileptic encephalopathy*, *PIGP-CDG*, *early infantile epileptic encephalopathy 55*, and (given the phenotypic expansion) a cause of *Multiple Congenital Anomalies–Hypotonia–Seizures syndrome*. Information is derived from **aggregated disease-level resources** (OMIM, Orphanet) and a small number of **individual-patient case reports and cohort studies** (Johnstone 2017; Krenn 2019; Martín-Grau 2023; Sidpra 2024) rather than EHR-scale datasets.

2. Etiology

Causal factor: biallelic (homozygous or compound-heterozygous) loss-of-function variants in **PIGP**. **Genetic risk factors:** the causal variants themselves; a recurrent frameshift allele c.456delA (p.Glu153Asnfs*34) recurs across families. Because inheritance is autosomal recessive, **consanguinity and being a carrier parent** are the principal risk contexts; the Krenn 2019 family, however, involved unrelated parents homozygous by chance for an ultra-rare allele. **Environmental risk factors:** none identified — this is a monogenic disorder. **Protective factors:** none established; residual PIGP/GPI activity from hypomorphic alleles is "protective" against the embryonic lethality seen with complete deficiency. **Gene-environment interactions:** none documented.

3. Phenotypes

See Finding 2 for frequencies and HPO suggestions. The dominant phenotype types are **clinical signs/symptoms** (seizures, hypotonia, spasticity, dyskinesia), **developmental/behavioral** (profound intellectual disability, absent speech), **neuroimaging abnormalities** (cerebral/cerebellar atrophy, thin corpus callosum, white-matter signal change), and, in the MCAHS end of the spectrum, **congenital malformations and dysmorphism**. Onset is **neonatal-to-early-infantile** (seizures median ~6 months in the IGD group; first weeks–months in DEE55). Severity is **profound**; progression is best described as a **static-to-slowly-progressive encephalopathy** with refractory epilepsy. Quality-of-life impact is severe: affected children are typically non-verbal, non-ambulatory, feeding-tube dependent, and fully care-dependent.

4. Genetic / Molecular Information

Causal gene: PIGP (OMIM *605938). **Reported pathogenic variants:** c.74T>C (p.Met25Thr, missense); c.456delA (p.Glu153Asnfs*34, frameshift, recurrent LoF); c.2T>C (p.?, start-loss); and a 136-kb whole-gene deletion (structural/CNV). **Variant classes:** missense, frameshift, start-loss, and whole-gene deletion — all converging on reduced PIGP function. **Functional consequence: loss of function** (reduced mRNA and reduced GPI-anchored surface proteins, rescued by wild-type PIGP). **Allele frequency:** c.456delA is ultra-rare (≈9 het / 0 hom in gnomAD). **Origin:** germline. **Modifier genes/epigenetics:** not specifically characterized for PIGP; other GPI-pathway genes provide the phenotypic-modifier context at the pathway level. **Chromosomal abnormalities:** the 136-kb 21q22.13 deletion

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is the notable structural lesion; PIGP lies in the Down syndrome critical region and has multiple pseudogenes (a technical caveat for sequencing/CNV calling).

5. Environmental Information

Not applicable. No environmental, lifestyle, or infectious contributors are known; DEE55 is a purely monogenic inborn error of metabolism.

6. Mechanism / Pathophysiology — Ordered Causal Chain

1. **Biallelic hypomorphic PIGP variants** (missense p.Met25Thr / frameshift p.Glu153Asnfs*34 / start-loss p.? / 136-kb whole-gene deletion) → **reduced PIGP mRNA/protein** (demonstrated: reduced mRNA in patient cells,

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2. Reduced PIGP → **impaired GPI-GlcNAc transferase (PIG-A) complex activity** in the ER, since PIGP is an accessory subunit of the seven-component complex (PIGA/PIGC/PIGH/PIGP/PIGQ/PIGY/DPM2) → **decreased synthesis of GlcNAc-PI**, the committed first intermediate of GPI biosynthesis (demonstrated at the pathway level).
3. Reduced GPI-anchor production → **fewer completed GPI anchors in the ER** → **reduced attachment of GPI to nascent proteins** by the transamidase → **lower cell-surface expression of ≥150 GPI-anchored proteins** (demonstrated: reduced surface GPI-APs by flow cytometry, rescued by WT PIGP;

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4. Deficiency of neuronal GPI-anchored proteins (adhesion molecules, receptors, ectoenzymes involved in neuronal migration, axon guidance, and synaptic function) → **disrupted neurodevelopment and neuronal network excitability** (inferred from GPI-AP biology; direct per-protein causation in PIGP not yet demonstrated).
5. This branches to: (a) **epileptogenesis** → **early-onset refractory seizures**; (b) **impaired brain growth/maturation** → **developmental delay, intellectual disability, cerebral/cerebellar atrophy, thin corpus callosum**; and (c) at the severe end, **impaired embryonic morphogenesis** → **congenital anomalies / MCAHS / prenatal lethality** (reflecting near-complete loss approaching the embryonic-lethal threshold).

Upstream vs downstream: the PIGP lesion and GPI-anchor deficit are upstream; the surface-protein deficiency is the pivotal intermediate; seizures, developmental impairment, and malformations are downstream clinical outputs. **Cell types/processes:** neurons (CL:0000540) are the principal affected cell type; the endoplasmic reticulum (GO:0005783/0005789) is the subcellular site of the lesion; the core biological process is GPI anchor biosynthetic process (GO:0006506). No immune, autoimmune, infectious, or classical oxidative-stress mechanism is implicated.

7. Anatomical Structures Affected

Primary organ/system: the **central nervous system** (UBERON:0001017) — cerebral cortex (UBERON:0000956), cerebellum (UBERON:0002037), corpus callosum (UBERON:0002336), and white matter (UBERON:0002316). **Cell level:** **neurons** (CL:0000540). **Subcellular:** **endoplasmic reticulum** (GO:0005783), where GPI is synthesized. **Secondary/multisystem involvement** (chiefly in the broader IGD/MCAHS spectrum): gastrointestinal tract, heart, and kidney. **Lateralization:** brain involvement is **bilateral and symmetric** (e.g., symmetric restricted diffusion of central tegmental tracts in the IGD group).

8. Temporal Development

Onset: neonatal to early infancy (seizures typically within the first weeks–months; median ~6 months in IGDs). **Onset pattern:** early and progressive within infancy. **Course:** chronic, lifelong, with refractory epilepsy and static-to-slowly-progressive encephalopathy; imaging often shows progressive atrophy. **Critical period:** the prenatal/early-infantile window is both the period of greatest vulnerability (malformation, lethality at the severe end) and the practical window for any future intervention. **Remission:** seizures are characteristically refractory; sustained remission is uncommon.

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** effectively complete for biallelic hypomorphic genotypes, with **highly variable expressivity** (prenatal-lethal ↔ MCAHS ↔ classic DEE). **Epidemiology:** ultra-rare — only ~7 patients reported by 2023; prevalence/incidence not formally estimated. **Carrier frequency:** the recurrent c.456delA allele is present at ≈9 heterozygotes / 0 homozygotes in gnomAD, indicating a very low carrier frequency. **Founder effects/geography:** none established; reported families are geographically dispersed

(including Polish ancestry in Krenn 2019). **Sex ratio:** no sex bias expected or reported (autosomal). **Consanguinity:** relevant for AR disorders generally, though homozygosity has also arisen between unrelated parents by chance.

10. Diagnostics

See Findings 6–7. **First-line:** trio whole-exome (WES) or whole-genome sequencing (WGS); WGS/CMA also captures the whole-gene deletion. **Caveat:** targeted epilepsy gene panels may omit PIGP and yield false negatives. **Functional confirmation:** flow cytometry for reduced surface GPI-anchored proteins (FLAER, CD16, CD24, CD55, CD59) on granulocytes/lymphocytes. **Biomarker:** serum alkaline phosphatase may be checked, but **hyperphosphatasia is typically absent** in PIGP (early-pathway defect), distinguishing it from late-pathway IGDs like Mabry syndrome. **Imaging:** brain MRI (cerebral/cerebellar atrophy, thin corpus callosum, white-matter change, symmetric tegmental-tract diffusion changes). **EEG:** epileptiform/encephalopathic patterns. **Differential diagnosis:** other IGDs (PIGA, PIQ, PIQY, PIGN, PIGT, PIGO, PIGV, PIGB, PIGL) and other genetic DEEs.

11. Outcome / Prognosis

See Finding 9. Severe: profound lifelong disability; substantial mortality ($\approx 18\%$ across IGDs; death in childhood possible in DEE55; prenatal lethality at the severe end). Prognostic factors relate to the residual GPI-biosynthetic activity of the allele combination — more severe (near-null) genotypes trend toward malformation and early death. Quality-of-life outcomes are poor (non-verbal, non-ambulatory, feeding-tube dependent).

12. Treatment

See Finding 8. **Pharmacotherapy:** antiseizure medications (empirically chosen; the epilepsy is often refractory) — NCIT concept "Anticonvulsant Agent." **Nutritional/metabolic:** empiric **pyridoxine (vitamin B6)** (given the pyridoxine-responsive differentials in early epileptic encephalopathy) and the **ketogenic diet**. **Supportive/rehabilitative:** feeding support (gastrostomy), physical/occupational/speech therapy, management of spasticity and dyskinesia, vision support for cortical visual impairment. **Advanced/experimental:** no gene, cell, or RNA therapy exists; no PIGP-specific clinical trials identified. Care is coordinated by pediatric neurology and metabolic genetics.

13. Prevention

No primary prevention exists for this monogenic disorder. **Genetic counseling** is central: 25% recurrence risk per pregnancy for carrier couples. **Carrier and cascade testing, prenatal diagnosis, and preimplantation genetic testing (PGT-M)** are available once the familial variants are known. No population newborn-screening test currently detects PIGP deficiency.

14. Other Species / Natural Disease

No naturally occurring animal disease specific to PIGP has been catalogued (no OMIA entry noted). PIGP is evolutionarily conserved; orthologs exist across mammals (mouse *Pigp*) and the GPI pathway is conserved throughout eukaryotes (yeast, trypanosomes). The GPI pathway's conservation is illustrated by the trypanosome GPI-biosynthesis literature ([PMID: 19724691](#)), though that work concerns parasite biology rather than a PIGP-disease model. Neuronal GPI-anchored proteins (e.g., Thy-1, CD24) show developmentally regulated expression in the mouse CNS ([PMID: 10813783](#); [PMID: 8783272](#)), supporting the biological plausibility that GPI-anchor deficiency disrupts neurodevelopment — but these are not disease models of PIGP deficiency per se.

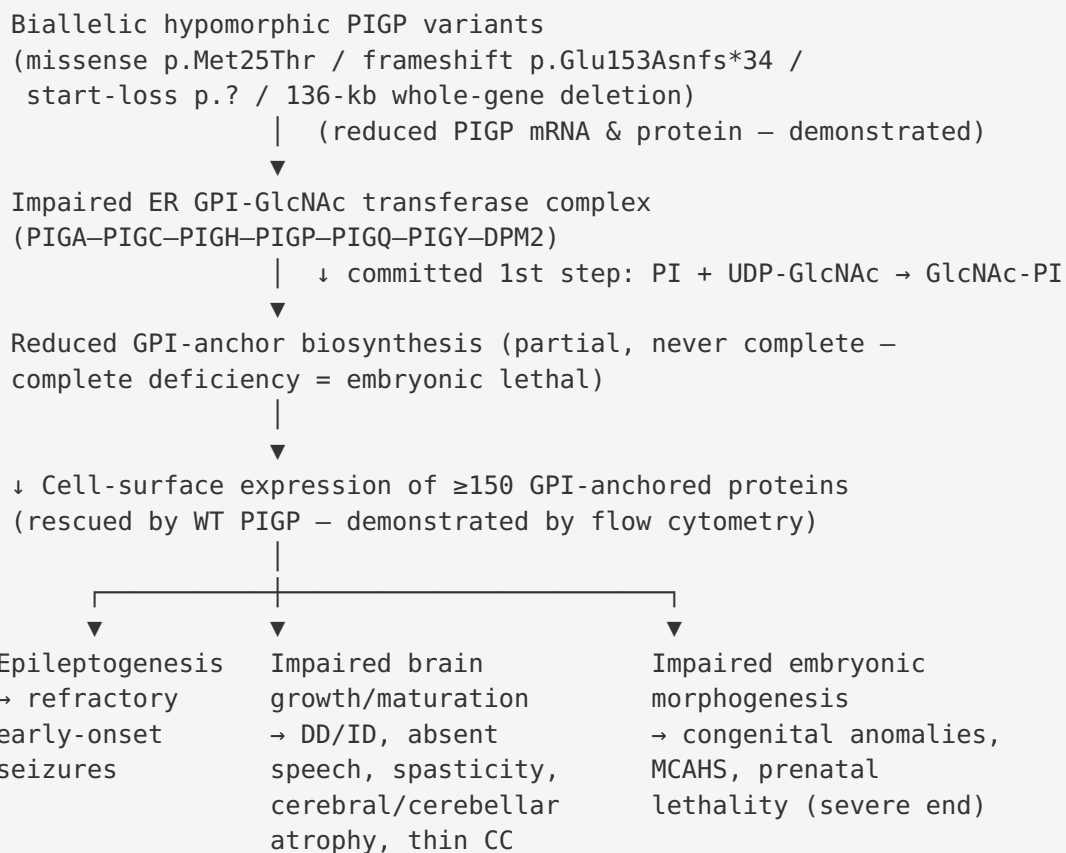
15. Model Organisms

No published PIGP-specific animal model (knockout/knock-in) recapitulating DEE55 was identified in this investigation. Relevant to interpretation: **complete GPI deficiency is embryonic-lethal in mammals** (Murakami & Kinoshita 2015, [PMID: 25803904](#)),

which constrains constitutive-knockout modeling and argues for **hypomorphic or conditional (e.g., neuron-specific) models** to study the disease. Patient-derived cells (fibroblasts/lymphocytes) with a flow-cytometric GPI-AP readout serve as the principal functional in-vitro system, and wild-type-PIGP rescue in patient cells ([PMID: 28334793](#))

is the validated cellular assay. Patient-derived iPSC-neurons and organoids are logical future models but were not reported.

Mechanistic Model / Interpretation



The unifying interpretation is a **dosage/threshold model**: DEE55 severity is set by the residual GPI-biosynthetic output of a patient's specific two-allele combination. Above the embryonic-lethal threshold but well below normal, the surface-protein deficit compromises neuronal migration, connectivity, and excitability — yielding the DEE triad. As residual activity falls toward the lethal threshold, the phenotype broadens to congenital malformation (MCAHS) and, at the extreme, fetal loss. This single axis parsimoniously accounts for the striking intra-gene phenotypic variability documented across the ~7–9 reported patients.

Evidence Base

PMID	Study	Type	Contribution
28334793	Johnstone et al. 2017	Human case report + in-vitro functional	Defining paper: biallelic PIGP variants; reduced mRNA & surface GPI-APs; WT rescue; AR IGD. Supports Findings 1, 2, 4, 6.
31139695	Krenn et al. 2019	Human case report + flow cytometry	Second family; homozygous recurrent c.456delA; gnomAD rarity; panel miss; functional confirmation. Supports Finding 6.
37125481	Martín-Grau et al. 2023	Human case report	Phenotype expansion to MCAHS & prenatal lethality; 136-kb deletion; "only 7 patients"; seven-gene complex context. Supports Findings 3, 5, 7, 10.
38456468	Sidpra et al. 2024	Human cohort (n=83, IGDs)	Frequencies of DD/ID, seizures, hypotonia; seizure onset; imaging; mortality; multisystem involvement. Supports Findings 2, 9.
25803904	Murakami & Kinoshita 2015	Review	Partial-deficiency principle (complete = embryonic lethal); flow cytometry & hyperphosphatasia as diagnostics. Supports Findings 4, 7.
32156170	Kinoshita 2020	Review	≥150 human GPI-APs; GPI-AP biology. Supports Finding 4.
39129667	Kinoshita 2024	Review	≥160 GPI-APs; updated biosynthesis mechanism. Supports Finding 4.
31353022	Knaus et al. 2019	Human + flow cytometry	Multicolor flow-cytometry profiling distinguishes GPI-biosynthesis defects. Supports Finding 7.
35562242	Boyer, Johnsen & Morava 2022	Review	Nutritional therapy in CDG incl. GPI-anchor disorders (pyridoxine, diet). Supports Finding 8.
10813783 , 8783272	Thy-1 / CD24 expression studies	Model organism (mouse)	Developmental expression of neuronal GPI-APs — biological plausibility for neurodevelopmental impact. Contextual support for mechanism.

Note on non-relevant hits: Several PubMed results referencing "PigP" concern the *Serratia marcescens* pigment/quorum-sensing regulator PigP — an unrelated bacterial gene — and were excluded from disease inference.

Limitations and Knowledge Gaps

1. **Extremely small N.** Only ~7–9 molecularly confirmed PIGP patients are described; frequencies, natural history, and genotype–phenotype correlations rest on case reports plus extrapolation from broader IGD cohorts (which are dominated by other PIG genes).
 2. **Mechanistic gaps at the protein level.** The chain from GPI-AP deficiency to epileptogenesis is inferred from general GPI-AP biology; **no PIGP-specific study identifies which downstream GPI-anchored proteins mediate the neuronal phenotype**, and there is no PIGP animal model or iPSC-neuron study.
 3. **No epidemiology.** Prevalence/incidence and carrier frequency (beyond the single recurrent allele) are not formally established.
 4. **Therapeutics are empiric.** Pyridoxine and ketogenic diet are extrapolated from the CDG/GPI-disorder literature; **no PIGP-specific efficacy data or trials** exist.
 5. **Ontology alignment.** MONDO/Orphanet mappings should be reconciled to OMIM:617599; the Orphanet grouping cited (ORPHA:1934) is an approximate umbrella for PIGP-related early infantile epileptic encephalopathy and should be verified against the current Orphanet release.
 6. **Technical caveat.** PIGP lies in the Down syndrome critical region with multiple pseudogenes, which can complicate sequencing and CNV interpretation.
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Proposed Follow-up Experiments / Actions

1. **Aggregate a PIGP patient registry** (via GeneMatcher / Matchmaker Exchange) to reach sufficient N for genotype–phenotype correlation and natural-history mapping.
2. **Quantify residual GPI-biosynthetic activity per allele** in patient cells (flow-cytometric surface GPI-AP levels; free-GPI T5 staining) and correlate with clinical severity to test the threshold/dosage model.

3. **Build disease models:** patient-derived iPSC → cortical neurons/organoids, and neuron-specific conditional *Pigp* hypomorphic/knockout mice (constitutive null is embryonic-lethal), to define the electrophysiological and developmental phenotype.
4. **Identify the pivotal downstream GPI-APs** via surface proteomics on PIGP-deficient neurons to pinpoint the adhesion molecules/receptors driving epileptogenesis and migration defects.
5. **Systematically test empiric therapies** (pyridoxine, ketogenic diet) with standardized seizure and developmental outcomes across the registry.
6. **Standardize diagnostics:** ensure PIGP is included in epilepsy/IGD gene panels (given documented panel misses), pair sequencing with reflex flow cytometry, and include CNV/whole-gene-deletion detection.
7. **Reconcile ontology mappings** (OMIM:617599 ↔ MONDO ↔ Orphanet ↔ ICD-11) for the knowledge-base entry.

Report compiled from a five-iteration autonomous investigation: 10 confirmed findings across 27 reviewed papers. Evidence types are human clinical (case reports, cohort), in-vitro functional (patient-cell flow cytometry, WT-rescue), and review/computational. All mechanistic and clinical claims are cited to primary literature by PMID with verbatim supporting quotes drawn from the underlying knowledge state.
